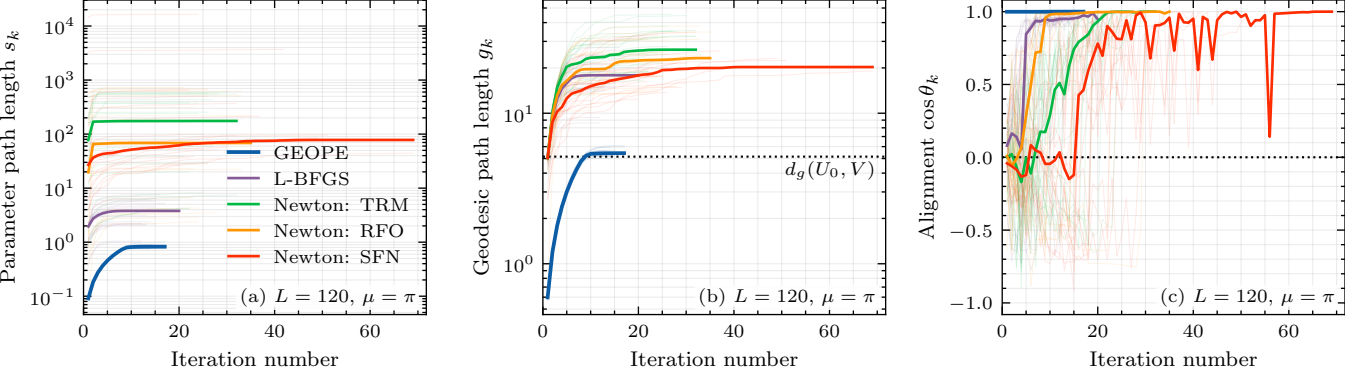

Parameter path length s_k
10^4
10^3
10^2
10^1
10^0
10^{-1}
0
20
40
60
Iteration number
GEOPE
L-BFGS
Newton: TRM
Newton: RFO
Newton: SFN
(a) $L = 120$, $\mu = \pi$
Geodesic path length g_k
10^1
10^0
$d_g(U_0, V)$
Iteration number
(b) $L = 120$, $\mu = \pi$
Alignment $\cos\theta_k$
1.0
0.5
0.0
−0.5
−1.0
Iteration number
(c) $L = 120$, $\mu = \pi$